AS MATHEMATICS - PURE 1 - DAY 12

Stationary points

Turning points and optimisation

YOUR AIM TODAY

Find, classify and interpret stationary points.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

The ideas you need

At a stationary point, the tangent is horizontal, so $dy/dx=0$.

- Solve $dy/dx=0$ to find stationary x-values.
- Substitute into y to find coordinates.
- Second derivative positive: local minimum; negative: local maximum.
- A sign change + to - is a maximum; - to + is a minimum.

MEMORY RULE

Differentiate, equal zero, solve, substitute, classify.

Classify

WORKED STEPS

$y=x^2-4x+1$. $dy/dx=2x-4=0$ gives $x=2$; $y=-3$. $d^2y/dx^2=2>0$, so minimum $(2,-3)$.

Cubic

WORKED STEPS

$y=x^3-3x$: $dy/dx=3x^2-3=0$ gives $x=\pm 1$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find stationary point of $y=x^2+6x+2$.	
2. Classify it.	
3. Find stationary x-values of $y=x^3-12x$.	
4. Find and classify both stationary points in Q3.	
5. Find minimum value of $x^2-8x+20$.	
6. Rectangle perimeter is 20. If sides are x and $10-x$, find maximum area.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	dy/dx=2x+6=0, x=-3; y=-7.
2	d2y/dx2=2>0, so a minimum.
3	3x^2-12=0, so x=+-2.
4	At x=-2, y=16 and second derivative -12: maximum. At x=2, y=-16 and second derivative 12: minimum.
5	Stationary x=4; minimum value 4.
6	A=x(10-x)=10x-x^2. A'=10-2x=0 gives x=5; maximum area 25.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.